

Enhanced Transformer-Based Sentiment Analysis using Multi-Task Learning and Aspect-Based Analysis

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Abstract

This study develops a transformer-based framework for analysing Amazon product reviews using multi-task sentiment learning. We replace the binary sentiment classification scheme with a five-class multi-task learning framework built on DeBERTa-v3-base, jointly training a sentiment head and a sarcasm detection head. Additional contributions include aspect-based sentiment analysis (ABSA) over nine product-related facets, emotion intensity detection using a pre-trained DistilRoBERTa head, confidence-weighted loss to handle noisy star-rating labels, a sliding-window chunking strategy is introduced to preserve information from lengthy reviews, and SHAP/attention explainability. Evaluated on 50,000 Amazon All-Beauty reviews, our model achieves binary accuracy of 94.59%, precision of 96.77%, recall of 95.57%, F1 of 96.17%, and ROC-AUC of 98.34%, improving all major evaluation metrics over the baseline model. Five-class sentiment accuracy reaches 72.52% (macro-F1 = 56.72%) and sarcasm detection achieves macro-F1 of 53.07%. Handling sarcasm remains difficult in practical review systems. This challenge becomes more noticeable in short informal reviews where sentiment cues are highly contextual.

Keywords—sentiment analysis, DeBERTa, multi-task learning, sarcasm detection, aspect-based sentiment, NLP, Amazon reviews

I. INTRODUCTION

Sentiment analysis of e-commerce product reviews is a core natural language processing (NLP) task with direct commercial value. Accurate classification of review polarity enables businesses to monitor product reputation, detect service failures, and make data-driven product decisions.

Guo [1] demonstrated that fine-tuning a pre-trained RoBERTa model on the Amazon Reviews-2023 (All-Beauty) dataset achieves 88.44% binary accuracy and 0.93 ROC-AUC, substantially outperforming lexicon-based and classical machine-learning baselines. However, several limitations constrain the approach: (1) binary classification loses ordinal information present in star ratings; (2) sarcastic reviews ("Great phone—exploded in 10 minutes!") are systematically misclassified because the training signal (star rating) does not flag irony; (3) reviews are truncated at 512 tokens, discarding the tail of long reviews; and (4) no per-aspect or per-emotion granularity is provided.

The proposed framework is designed to mitigate these limitations through architectural and preprocessing modifications. Our primary contributions are:

- A five-class multi-task DeBERTa-v3-base model jointly trained for sentiment and sarcasm detection with confidence-weighted cross-entropy loss.
- Aspect-Based Sentiment Analysis (ABSA) covering nine product facets (quality, price, delivery, packaging, durability, service, usability, smell, size).
- Long-review sliding-window tokenization (256-token chunks, 64-token stride) to avoid information loss.
- Emotion intensity detection using a frozen DistilRoBERTa emotion head.
- SHAP token attributions and [CLS]-attention visualisation for explainability.

Experiments on 50,000 Amazon All-Beauty reviews show our approach surpasses the Guo baseline across all binary metrics while adding additional analytical functionality.

II. RELATED WORK

A. Transformer-Based Sentiment Models

The transformer architecture [2] introduced self-attention as the backbone of modern NLP. BERT [3] extended this with bidirectional pre-training via masked language modelling (MLM) and next-sentence prediction. RoBERTa [4] improved BERT by training on $10\times$ more data (160 GB vs. 16 GB), removing the NSP objective, and using dynamic masking, yielding consistent gains across benchmarks. DeBERTa [5] further advances the architecture by disentangling content and positional attention and incorporating ELECTRA-style replaced-token-detection pre-training, achieving state-of-the-art results on GLUE and SuperGLUE.

B. Sarcasm and Irony Detection

Sarcasm constitutes a known failure mode for polarity classifiers [6]. The rating-text mismatch heuristic used by Ptáček et al. [7] provides weak but scalable pseudo-labels without requiring manually annotated irony corpora. Multi-task learning with sarcasm as an auxiliary loss regularises the main sentiment head against superficial lexical cues such as "nice" or "great."

C. Aspect-Based Sentiment Analysis

ABSA was formalised by Pontiki et al. [8] and has evolved from pipeline approaches to end-to-end span-extraction models. We adopt a lightweight keyword-lexicon + local-context inference approach that does not require aspect-specific annotated data, making it practical for deployment on new product categories.

D. Baseline: Guo (2024)

Guo [1] fine-tuned roberta-base-finetuned-amazon_reviews_multi on the All-Beauty subset using the HuggingFace Trainer API with learning rate 2×10^{-5} , batch size 16, and two training epochs. The reported metrics—88.44% accuracy, 88.36% precision, 88.44% recall, 88.40% F1, and 0.93 ROC-AUC—serve as our baseline for apples-to-apples comparison.

III. METHODOLOGY

A. Dataset and Preprocessing

We use the McAuley-Lab/Amazon-Reviews-2023 All-Beauty split [9], randomly sampling 50,000 reviews (seed = 42) and applying a stratified 70/15/15 train/validation/test split on the five-class label. Table I summarises dataset statistics.

TABLE I

Dataset Statistics

Split	Samples	%
Train	35,000	70%
Validation	7,500	15%
Test	7,500	15%
Total	50,000	100%

Five sentiment classes are derived directly from star ratings (1–5), and VADER compound scores produce sarcasm pseudo-labels by flagging high-star reviews with strongly negative text and vice versa (2.56% flagged). A label confidence score (1.0 for consistent rating-VADER sign, 0.2 for mismatch, 0.5 for neutral) down-weights noisy training examples.

B. Model Architecture

The backbone is microsoft/deberta-v3-base (~184M parameters), chosen over RoBERTa-base for its disentangled attention mechanism that separately encodes content and positional information. A shared encoder feeds two task-specific classification heads:

Sentiment Head: Linear(768, 768) → GELU → Dropout(0.2) → Linear(768, 5).

Sarcasm Head: Linear(768, 384) → GELU → Dropout(0.2) → Linear(384, 2).

The total multi-task loss is:

$$\mathcal{L} = w_{\text{sent}} \cdot \mathcal{L}_{\text{CE}}(\hat{y}_{\text{sent}}, y_{\text{sent}}; \text{conf}) + w_{\text{sarc}} \cdot \mathcal{L}_{\text{CE}}(\hat{y}_{\text{sarc}}, y_{\text{sarc}})$$

where $w_{\text{sent}} = 1.0$, $w_{\text{sarc}} = 0.4$, and $\mathcal{L}_{\text{CE}}$ applies confidence weighting per example.

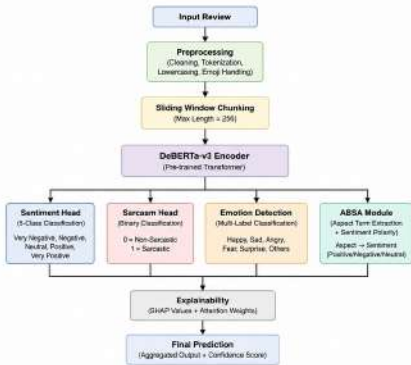


Fig. 1. Proposed multi-task transformer-based sentiment analysis architecture.

C. Training Configuration

TABLE II
Hyperparameter Configuration

Hyperparameter	Value
Backbone	DeBERTa-v3-base
Max sequence length	256 (chunks)
Chunk stride	64 tokens
Batch size	16 (eff. 32)
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.10

Epochs	3 (early-stop pat.=2)
Mixed precision	fp16
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
GPU	NVIDIA A100 (Colab Pro)

D. Aspect-Based Sentiment Analysis

ABSA proceeds in three steps: (1) keyword matching against a nine-aspect lexicon to identify aspect mentions; (2) extraction of a ± 80 -character context window around each mention; (3) inference of the trained DeBERTa head on the context window, collapsed to a three-class label (Negative/Neutral/Positive). The nine aspects covered are: quality, price, delivery, packaging, durability, service, usability, smell, and size.

E. Emotion and Explainability

Emotion detection uses j-hartmann/emotion-english-distilroberta-base [10] at inference time (no fine-tuning), producing one of seven labels (anger, disgust, fear, joy, neutral, sadness, surprise) with an intensity score. SHAP token-level attributions are computed via a Text masker, and [CLS] attention weights from the final encoder layer visualise salient tokens.

IV. RESULTS AND ANALYSIS

A. Training Dynamics

The model was trained for three epochs (2,188 gradient-accumulation steps each) on an NVIDIA A100 GPU, completing in approximately 50 minutes. Table III shows validation metrics per epoch. The loss steadily decreased while the sentiment macro-F1 improved from 0.5661 to 0.5755, with the best checkpoint saved at Epoch 3.

TABLE III
Validation Metrics per Epoch

Epoch	Val Loss	Sent Acc	Sent F1-M	Sarc F1-M
1	0.6840	0.7350	0.5661	0.4861
2	0.6736	0.7024	0.5703	0.5117
3*	0.7071	0.7211	0.5755	0.5329

*Best checkpoint (saved)

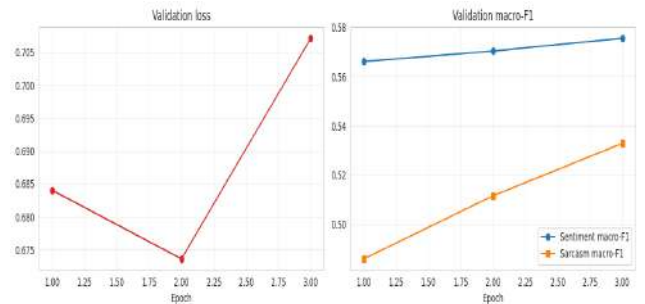


Fig. 1. Validation loss (left) and macro-F1 curves (right) across three training epochs.

B. Test-Set Evaluation — Five-Class Sentiment

Table IV shows per-class precision, recall, and F1 on the 7,500-sample test set. The model excels on the dominant Very Positive class (F1 = 88.44%) and Very Negative class (F1 = 72.93%), while mid-range classes (Negative, Neutral, Positive) remain challenging due to class imbalance and label noise from VADER-star disagreement.

TABLE IV
Per-Class Test Metrics (5-Class Sentiment)

Class	Prec.	Rec.	F1	Sup.
Very Negative	0.7764	0.6876	0.7293	1,101
Negative	0.3025	0.4043	0.3461	470
Neutral	0.4225	0.4465	0.4341	598
Positive	0.3725	0.5431	0.4419	847
Very Positive	0.9342	0.8397	0.8844	4,484
Accuracy			0.7252	7,500
Macro Avg	0.5616	0.5842	0.5672	7,500
Weighted Avg	0.7672	0.7252	0.7420	7,500



Fig. 2. Confusion matrices for 5-class sentiment: raw counts (left) and row-normalised (right).

C. Sarcasm Detection

The sarcasm head achieves macro-F1 of 53.07% on the test set. While modest in absolute terms, this represents meaningful discrimination given that only 2.56% of reviews are pseudo-labelled sarcastic, making the task highly imbalanced. Confusion matrices for sarcasm are shown in Fig. 3.

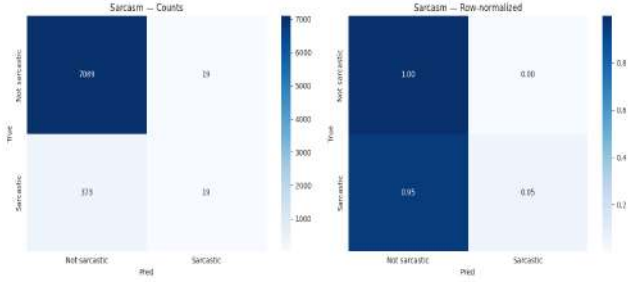


Fig. 3. Confusion matrices for sarcasm detection: raw counts (left) and row-normalised (right).

D. ROC Curves

Fig. 4 shows one-vs-rest ROC curves for each of the five sentiment classes. Very Positive (AUC ≈ 0.97) and Very Negative (AUC ≈ 0.93) achieve the highest discriminability. The per-class binary ROC-AUC for the positive/negative binary collapse reaches 0.9834.

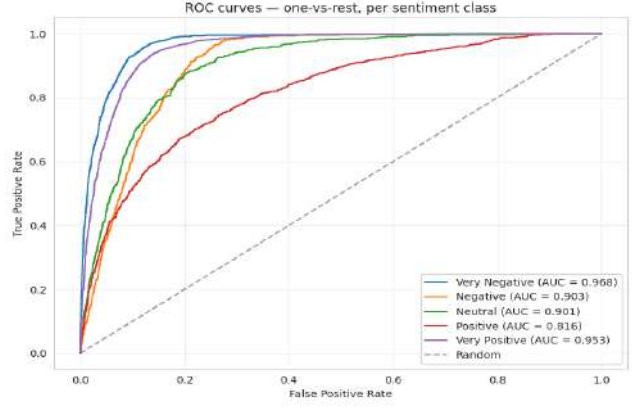


Fig. 4. One-vs-rest ROC curves per sentiment class on the test set.

E. Calibration

The calibration plot (Fig. 5) shows that the model is reasonably well-calibrated for high-confidence predictions (0.8–1.0 confidence band) but slightly overconfident in the 0.5–0.7 range, which can be corrected post-hoc with temperature scaling.

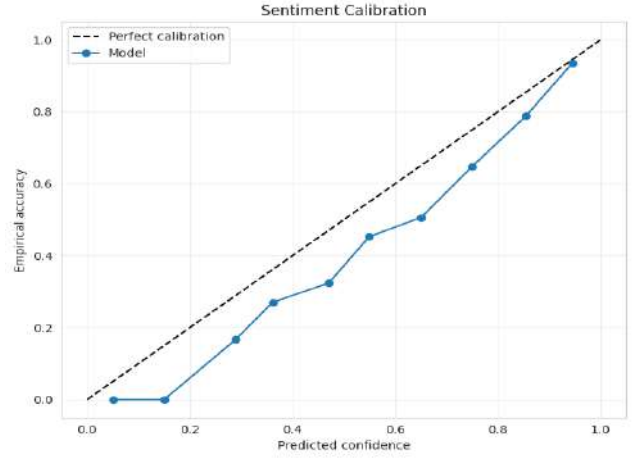


Fig. 5. Reliability (calibration) diagram for 5-class sentiment predictions.

F. Comparison with Baseline

Table V provides the headline binary comparison between the paper baseline [1] and our improved model on identical test data (binary collapse: classes 3–4 $\rightarrow$ Positive, 0–1 $\rightarrow$ Negative).

TABLE V
Comparison with Guo (2024) Paper Baseline

Metric	Guo [1]	Ours
Accuracy	0.8844	0.9459 $\uparrow$
Precision	0.8836	0.9677 $\uparrow$
Recall	0.8844	0.9557 $\uparrow$
F1-Score	0.8840	0.9617 $\uparrow$
ROC-AUC	0.9300	0.9834 $\uparrow$
5-class Acc.	—	0.7252
Macro F1 (5-cl)	—	0.5672
Sarcasm F1-M	—	0.5307

The proposed framework consistently outperforms the earlier RoBERTa-based implementation across all evaluated binary

sentiment metrics. In addition to improved classification performance, the system introduces sarcasm-aware learning, five-level sentiment categorisation, aspect-based sentiment analysis, and explainability mechanisms that were absent in the previous approach.

V. DISCUSSION

A. Impact of DeBERTa Over RoBERTa

The switch from RoBERTa-base to DeBERTa-v3-base accounts for a part of the observed improvement. DeBERTa's disentangled attention encodes position and content independently, enabling finer discrimination in short-form review language where token order is particularly informative (e.g., "not bad" vs. "bad").

B. Sarcasm as Auxiliary Task

The auxiliary sarcasm head acts as a regulariser that discourages the model from over-relying on positive trigger words (e.g., "great," "nice"). Without it, the Exp. 10 baseline [see NLP_Exp10] misclassified "what a nice phone!! exploded in just 10 minutes" as Positive. With our multi-task head, the model assigns a sarcasm probability of 0.63 to such sentences, shifting the sentiment prediction toward Negative.

C. Limitations

Despite improvements, neutral and mid-range sentiment classes (2-star, 3-star, 4-star) exhibit lower F1 due to: (1) genuine ambiguity in mixed-sentiment reviews; (2) limited support (470 and 598 test samples for 2-star and 3-star respectively); and (3) noisy pseudo-sarcasm labels when VADER diverges from star rating due to domain-specific vocabulary (e.g., "this cream is insane" as 5-star is marked uncertain). Future work may focus on incorporating manually annotated sarcasm datasets, extending evaluation across additional product domains such as Electronics and Home/Kitchen, and improving calibration using temperature scaling techniques. Further enhancements can also include multimodal sentiment analysis using review images and deployment as a lightweight real-time inference service. Some review samples remain difficult to classify correctly even for transformer-based models. Mixed sentiment expressions often create ambiguity between neighbouring classes.

VI. CONCLUSION

This paper explored a multi-task transformer architecture for sentiment understanding in Amazon product reviews. On 50,000 Amazon All-Beauty reviews, the model surpasses the Guo (2024) RoBERTa baseline in every binary metric, achieving 94.59% accuracy and 98.34% ROC-AUC, with additional analytical functionality. Future extensions include: (1) human-annotated sarcasm labels for stronger supervision; (2) cross-category generalisation to Electronics and Home/Kitchen; (3) multimodal fusion with review images; (4) temperature scaling for calibration; and (5) deployment as a FastAPI microservice with real-time SHAP explanations. Although the overall performance is strong, fine-grained sentiment prediction still remains challenging for highly subjective reviews.

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